

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 30

WORLD CLIMATIC TYPES

- **THE HOT, WET EQUATORIAL CLIMATE**

CLIMATOLOGY MODULES

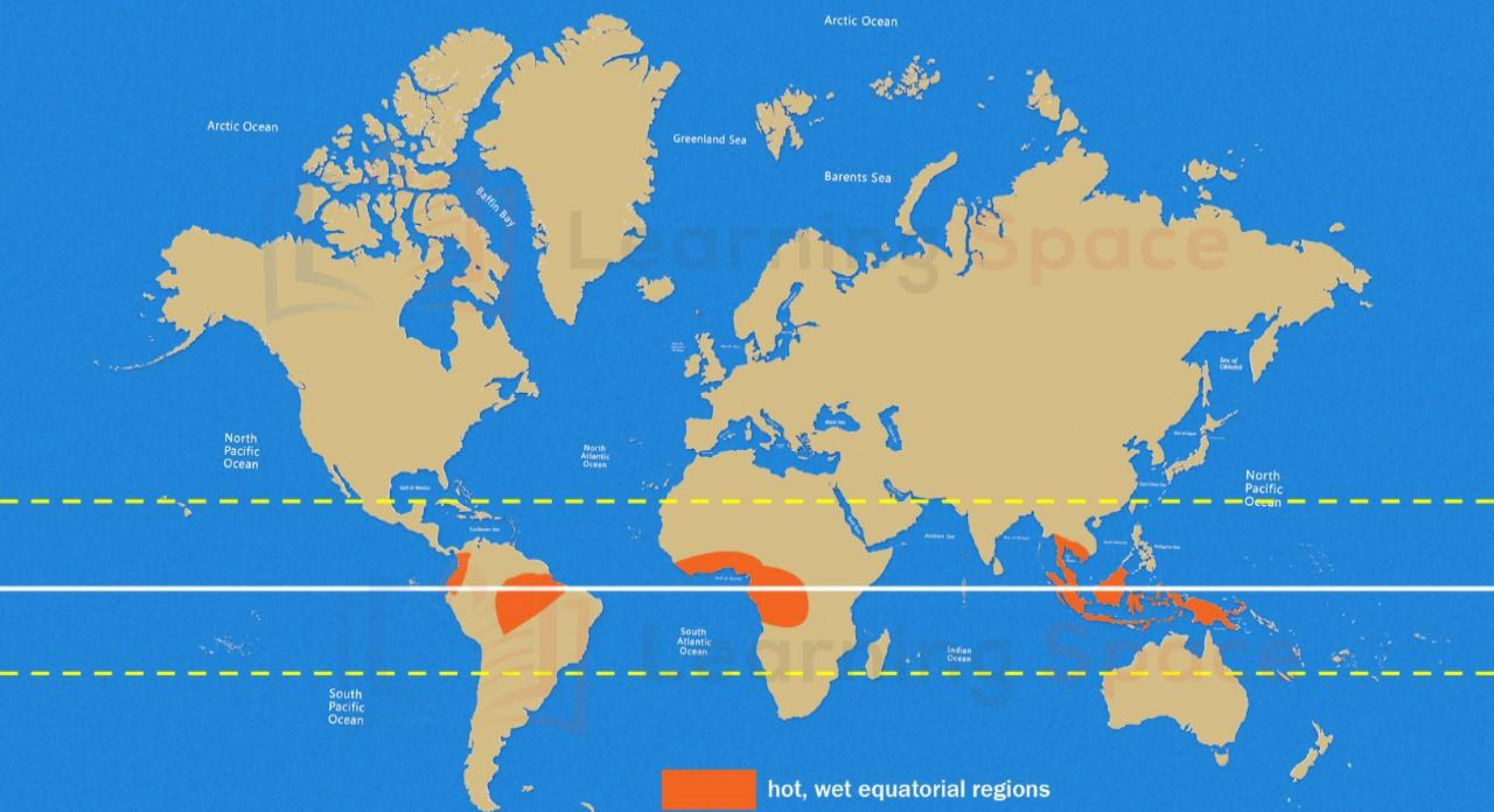
Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth — heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature — horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation — dew, frost, fog, mist and cloud; rainfall — types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate

WORLD CLIMATIC TYPES

1 THE HOT, WET EQUATORIAL CLIMATE DISTRIBUTION



- 1 The equatorial hot, wet climate is found between 5° and 10° north and south of the equator.
- 2 Its greatest extent is found in the lowlands of the Amazon, the Congo, Malaysia and the East Indies.
- 3 Further away from the equator, the influence of the on-shore Trade Winds, gives rise to modified type of equator climate with monsoonal influences.



WORLD CLIMATIC TYPES

1 THE HOT, WET EQUATORIAL CLIMATE

CLIMATE

a

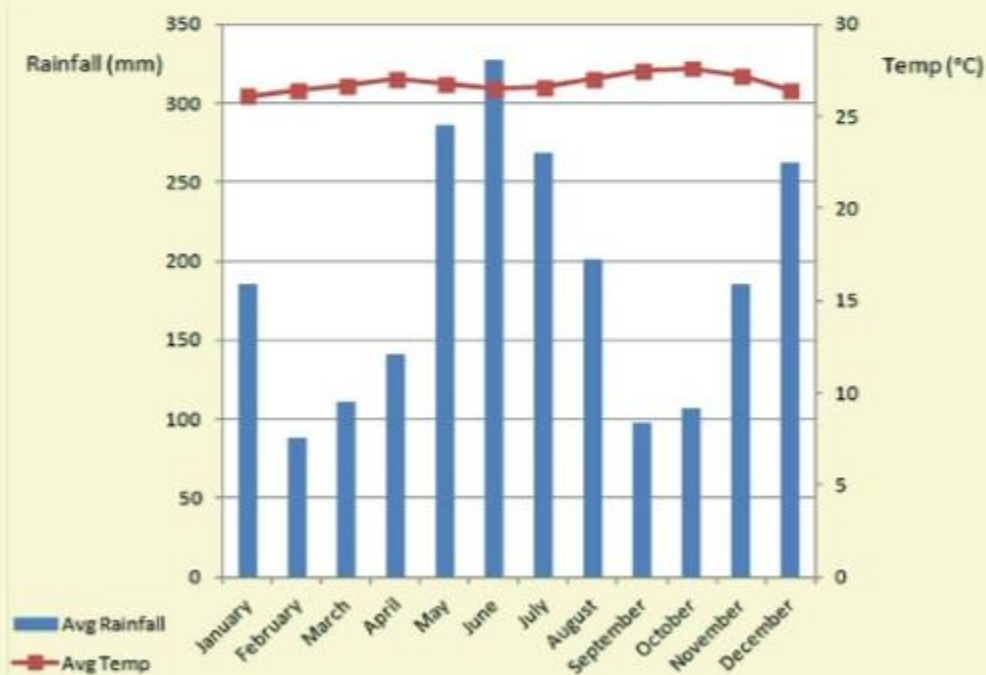
TEMPERATURE

1 The most outstanding feature of the equatorial climate is its great uniformity of the temperature throughout the year.

2 The mean monthly temperatures are always around 27°C with very little variation. There is no winter.

3 Cloudiness and heavy precipitation help to moderate the daily temperature, so that even at the equator the climate is not unbearable.

4 In addition, regular land and sea breezes assist in maintaining a truly equable climate.



WORLD CLIMATIC TYPES

1 THE HOT, WET EQUATORIAL CLIMATE

CLIMATE

b

PRECIPITATION

- 1 Precipitation is heavy, between **150 cm and 250 cm** and well distributed throughout the year.



- 2 There is **no month without rain** and a **distinct dry season** like those of the Savanna or the Tropical Monsoon Climates, **is absent**.
- 3 Instead, there are two periods of maximum rainfall in April and October.
- 4 The **double rainfall peaks coinciding with the equinoxes** are a characteristic feature of equatorial climates not found in any other types of climate.

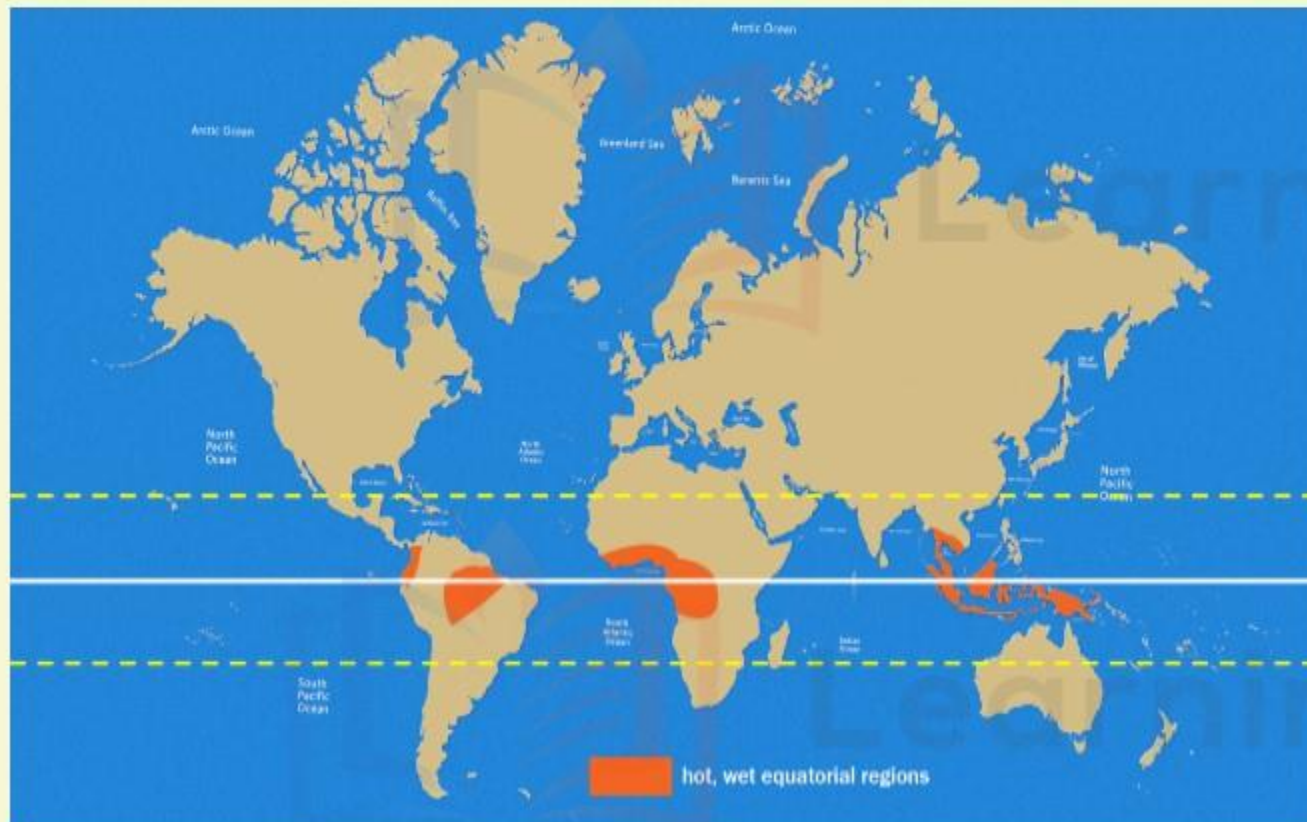
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THE HOT, WET EQUATORIAL CLIMATE

CLIMATE

b

PRECIPITATION



5

As one goes further north and south of the equator, particularly in coastal districts open to the influences of the trades, the tendency is towards a monsoonal pattern with the heaviest rainfall coming in the summer months

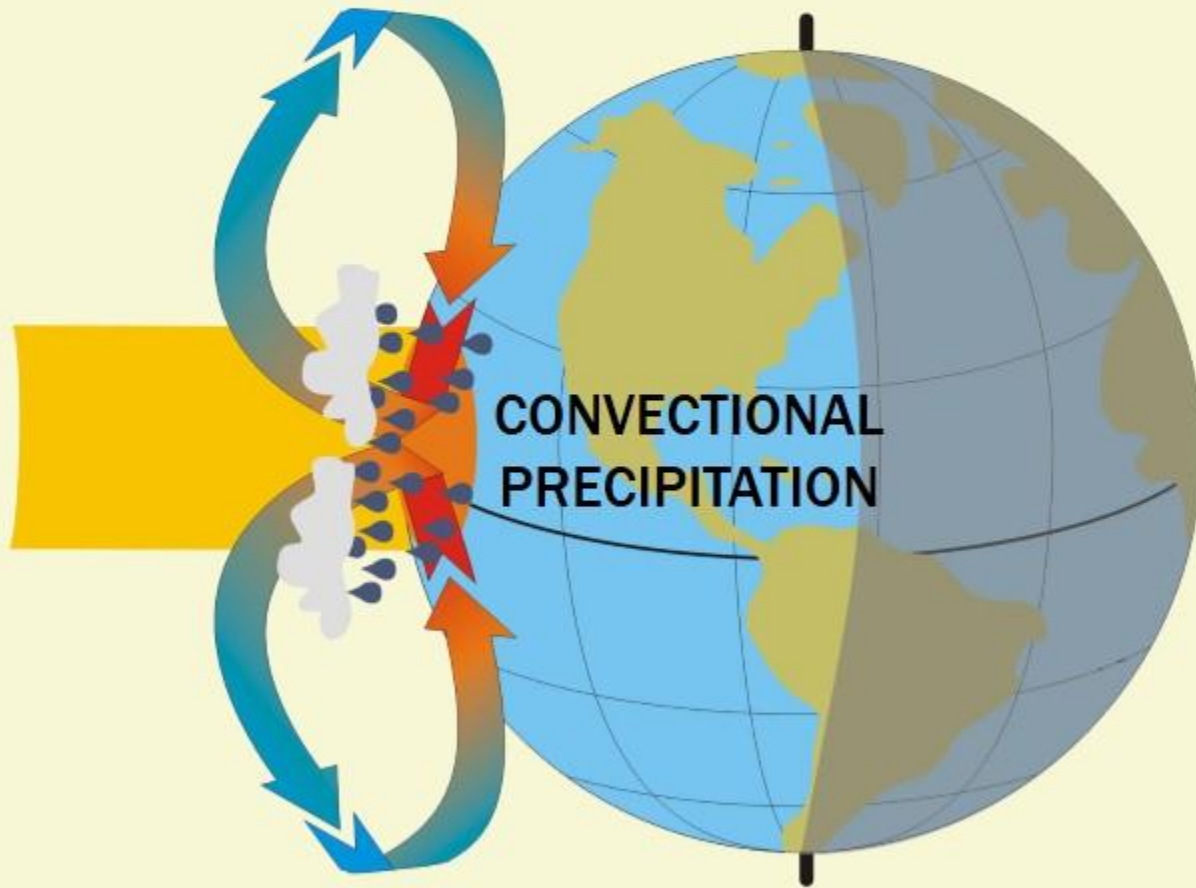
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Due to the great heat in the equatorial belt, mornings are bright and sunny. There is much evaporation and convectional air currents are set up due to the intense heating.

1 THE HOT, WET EQUATORIAL CLIMATE

CLIMATE

b PRECIPITATION



7 This is followed by **heavy downpours of convectional rain in the afternoons** from the towering cumulonimbus clouds.

8 **Thunder and lightning** often accompany the torrential showers and the amount of rainfall recorded in one single afternoon may be as much as the deserts receive for the entire year!

9 Besides the convectional rainfall, **mountainous regions** also experience much **orographic or relief rain**.

1 THE HOT, WET EQUATORIAL CLIMATE

CLIMATE

b PRECIPITATION



9 The relative humidity is constantly high (over 80 per cent) making one feel 'sticky' and uncomfortable.

10 The monotonous climate, oppressive and enervating, taxes one's mental alertness and physical capability, though along the coasts refreshing sea breezes do bring some relief.

1 THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

1 High temperature and abundant rainfall in the equatorial regions support a luxuriant type of vegetation the tropical rain forest.

2 In the Amazon lowlands, the forest is so dense and so complete in its vegetational extravagance that a special term 'selvas' is used.

3 Unlike the temperate regions, the growing season here is all the year round- seeding, flowering, fruiting, and decaying do not take place in a seasonal pattern, so some trees may be in flower while others only a few yards away may be bearing fruit.

4 There is neither drought nor cold to check growth in any part of the year

1 THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

The characteristic features of the equatorial vegetation may be summarized as follows:

a

A GREAT VARIETY OF VEGETATION

1 The equatorial vegetation comprises a multitude of evergreen trees that yield **tropical hardwood** e.g. mahogany, ebony, greenheart, cabinet woods and dyewoods.



2 There are smaller palm trees, **climbing plants like the lianas or rattan** which may be hundreds of feet long and **epiphytic and parasitic plants that live on other plants**.



1

THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

a

A GREAT VARIETY OF VEGETATION



EPIPHYTES



LIANAS

1 THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

b A DISTINCT LAYER ARRANGEMENT



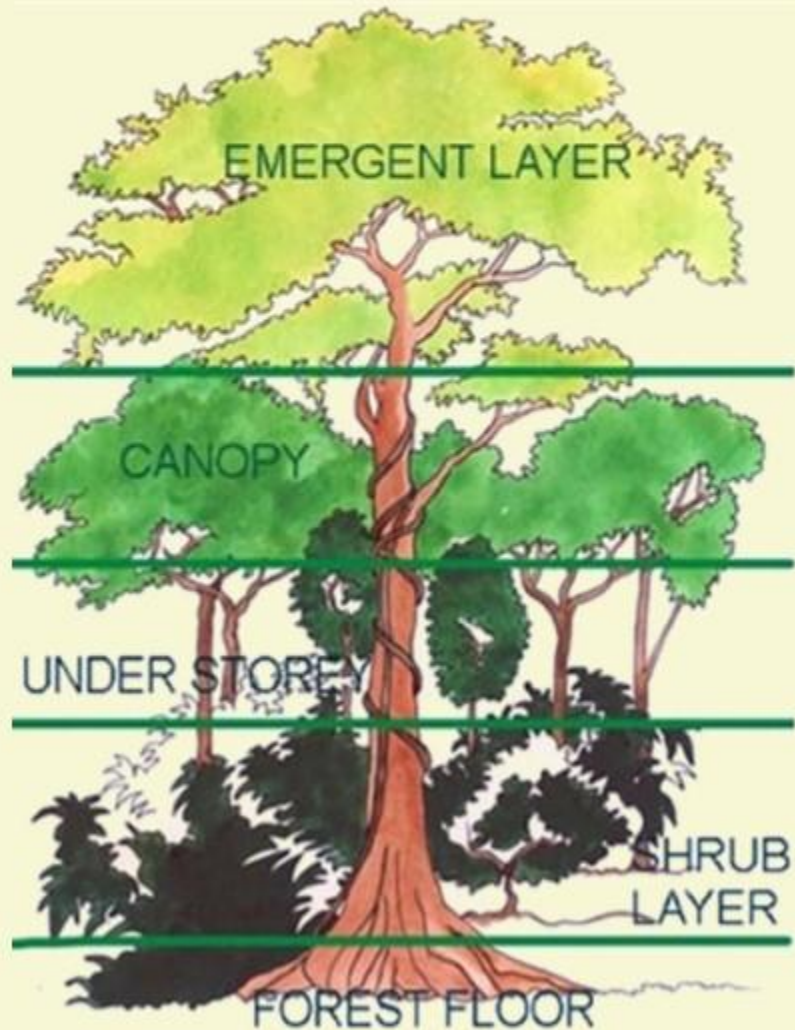
1 From the air, the tropical rain forest appears like a thick canopy of foliage, broken only where it is crossed by large rivers or cleared for cultivation.

2 All plants struggle upwards for sunlight resulting in a peculiar layer arrangement.

1 THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

b A DISTINCT LAYER ARRANGEMENT



3 The tallest trees attain a height of over 150 feet. Their slender trunks pierce skywards with wide-spread branches at the top.

4 The smaller tree beneath from the next layer and the ground is rooted with ferns and herbaceous plants which can tolerate shade.

5 Because the trees cut out most of the sunlight the undergrowth is not dense.

1 THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

b MULTIPLE SPECIES



1 Unlike the temperate forests, where only a few species occur in particular areas, the trees of the tropical rain forests **are not found in pure stands of a single species.**

2 This has made commercial exploitation of tropical timber a most difficult task.

3 Many of the tropical hardwoods do not float readily on water and makes this haulage an expensive matter. It is therefore not surprising that **many tropical countries are net timber importers!**

1 THE HOT, WET EQUATORIAL CLIMATE

EQUATORIAL VEGETATION

c FOREST CLEARINGS



1 Many parts of the virgin tropical rain forests have been cleared **either for lumbering or shifting cultivation.**

2 When these clearings are abandoned, **less luxuriant secondary forests, called belukar** in Malaysia, spring up.

1 THE HOT, WET EQUATORIAL CLIMATE

LIFE AND DEVELOPMENT IN THE EQUATORIAL REGIONS

1 The equatorial regions are generally sparsely populated.

2 In the forests, most primitive people live as hunters and collectors and the more advanced ones practise shifting cultivation.

3 Food is so abundant in such a habitat that many people worry very little about the life of the next day.

4 There are numerous animals, birds and reptiles that can be hunted too serve the needs of the community.

5 The ever-flowing rivers and streams provide an inexhaustible supply of fish that people spear or trap.

6 From the forest, they gather leaves, fruits, nuts and other forest products

1 THE HOT, WET EQUATORIAL CLIMATE

LIFE AND DEVELOPMENT IN THE EQUATORIAL REGIONS



7 With the coming of Europeans, many large plantations have been established especially in Java, Sumatra, Malaysia, West Africa and Central America.

8 The climate has proved to be very favourable for the cultivation of certain crops that are highly valued in the industrial West.

9 The most outstanding crop is natural rubber. Malaysia and Indonesia are its leading producers. The home country, Brazil exports practically no natural rubber.

1 THE HOT, WET EQUATORIAL CLIMATE

LIFE AND DEVELOPMENT IN THE EQUATORIAL REGIONS



10 The problems of tree diseases and the lack of commercial organisation of the Indians in the Amazon lowlands have brought about this unexpected 'shift' of rubber cultivation.

11 Another tropical crop that has achieved an amazing success is cocoa. It is most extensively cultivated in West Africa, bordering the Gulf of Guinea.

12 The two most important producers are Ghana and Nigeria. There is a keen demand for the crop and acreages are rapidly on the increase.

1 THE HOT, WET EQUATORIAL CLIMATE

LIFE AND DEVELOPMENT IN THE EQUATORIAL REGIONS



13 Most of the crop leaves West Africa for Europe or North America for the cocoa and chocolate industry.

14 From the same area another crop, oil palm, has done equally well and many countries outside Africa have now taken to its cultivation.

15 Other crops that have been found suitable for the hot, wet equatorial climate and are extensively cultivated are coconuts, sugar, coffee, tea, tobacco, spices, cinchona, bananas, pineapples and sago.

1 THE HOT, WET EQUATORIAL CLIMATE

FACTORS AFFECTING THE DEVELOPMENT OF EQUATORIAL REGIONS

a EQUATORIAL CLIMATE AND HEALTH



SYMPTOMS OF YELLOW FEVER

- 1 Under conditions of excessive heat and high humidity, man is subject to serious physical and mental handicaps.
- 2 He exposes himself to such dangers as sun-stroke and to such diseases as malaria and yellow-fever.
- 3 Consequently, his capacity for active work is greatly reduced and his resistance to diseases is much weakened.

1 THE HOT, WET EQUATORIAL CLIMATE

FACTORS AFFECTING THE DEVELOPMENT OF EQUATORIAL REGIONS

b PREVALENCE OF BACTERIA & INSECT PESTS



1 The hot wet climate which stimulates rapid plant growth, also encourages the spread of insects and pests.

2 As germs and bacteria are more easily transmitted through moist air, equatorial conditions are ideal for the survival of such organisms.

3 Insects and pest not only spread diseases but are injurious to crops They plague both men and animals.

1 THE HOT, WET EQUATORIAL CLIMATE

FACTORS AFFECTING THE DEVELOPMENT OF EQUATORIAL REGIONS

c JUNGLE HINDERS DEVELOPMENT

1 The jungle is so luxuriant that it is quite a **problem to clear a small patch of it and even more difficult to maintain it.**

2 Undergrowth spring up as soon as the shade trees are cut and unless they are weeded at regular intervals, they may **choke crops and overwhelm estates.**

3 In the same way, roads and railways constructed through the equatorial lands have to cut through forests, dense thickets and swamps.

4 Those who build and maintain such roads encounter wild animals, poisonous snakes and insects.

5 Once completed, they have to be maintained at a high cost.

6 Many remote parts of the Amazon Basin, the Congo and Borneo are without modern communication lines. **The rivers form the only natural highways.**

1 THE HOT, WET EQUATORIAL CLIMATE

FACTORS AFFECTING THE DEVELOPMENT OF EQUATORIAL REGIONS

d RAPID DETERIORATION OF TROPICAL SOIL



1 It is a misconception that tropical soils are rich.

2 The torrential down pours soon wash out most of the soil nutrients. The soil deteriorates rapidly with subsequent soil erosion and soil impoverishment.

3 One may quote the Indonesian island of Java as an exception, because of its rich volcanic ashes and the energetic local people, much progress has been made.

1 THE HOT, WET EQUATORIAL CLIMATE

FACTORS AFFECTING THE DEVELOPMENT OF EQUATORIAL REGIONS

e DIFFICULTIES IN LUMBERING AND LIVESTOCK FARMING



LUMBERING

- 1 Though the equatorial regions have great potential in timber resources, commercial extraction is difficult.
- 2 The trees do not occur in homogenous stands and the tropical hardwoods are sometimes too heavy to float in the rivers, even if these flow in the desired directions.
- 3 Livestock farming is greatly handicapped by an absence of meadow grass, even on the highlands. The few animals like bullocks or buffaloes are kept mainly as beasts of burden.

1 THE HOT, WET EQUATORIAL CLIMATE

FACTORS AFFECTING THE DEVELOPMENT OF EQUATORIAL REGIONS

e DIFFICULTIES IN LUMBERING AND LIVESTOCK FARMING



TSETSE FLY



NAGANA DISEASE

4 Their yield in milk or beef is well below those of the cattle in the temperate grasslands.

5 The grass is so tall and coarse that it is not nutritious.

6 In Africa, domesticated animals are attacked by **tsetse flies** that cause **nagana**, a deadly disease (otherwise known as Sleeping Sickness).

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